Jansky 2021 Neuroblastoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

13/14 samples (≥269 cells) intermix on one continuous manifold with no per-patient islands; the kNN graph is a single connected component (27,987 cells; one 129-cell spillover sample dropped). Same-sample-neighbour fraction 0.233 vs mixing null 0.111 — well-integrated across 13 patients.

Gate 2 — module geography → GO (non-branching)

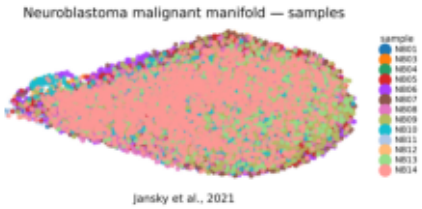
Gast/pSom decline together along pseudotime (ρ=−0.44/−0.14) and co-vary positively (r=+0.55); Crest is spatially anti-correlated (r=−0.72/−0.80) and rises to a terminal readout (ρ=+0.36; crest-high decile mean pt=0.72). Modules ride a dominant noradrenergic-differentiation axis — present, but not resolved into two competing arms.

Gate 3 — hub mutual exclusivity → NA

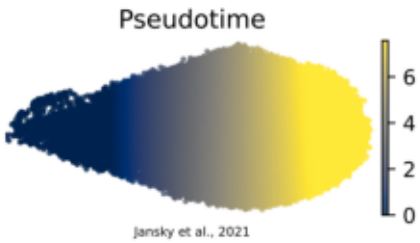
Single monotone differentiation axis — no bifurcation and no Gast/pSom-overlap hub, so there is no branch pair to test for competitive cross-repression.

STEP FIGURES

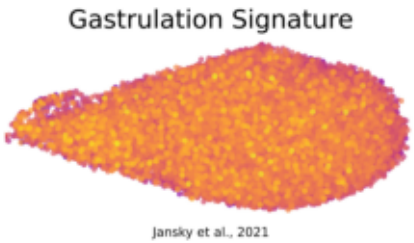
Manifold — samples



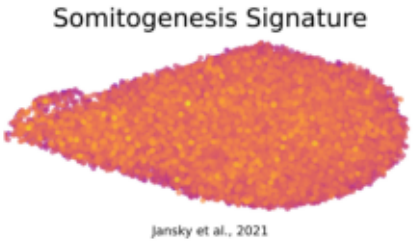
Pseudotime



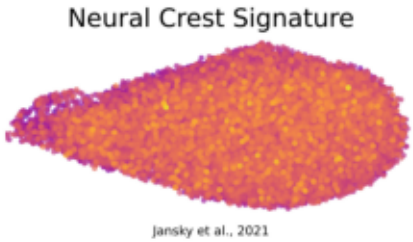
Gastrulation



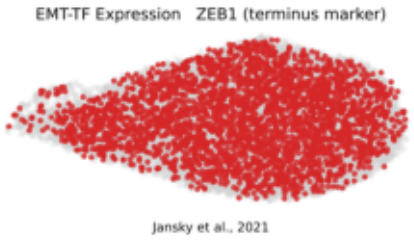
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Neuroblastoma (Jansky et al., 2021), 10x — childhood adrenal NB; malignant only: 28,116 load, 27,987 across 13 samples on one continuous manifold; the full compartment carries the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Jansky's manifold is a single noradrenergic-differentiation axis (progenitor → differentiating neuronal); embryonic modules ride it as a secondary Gast/pSom → terminal-Crest gradient, lacking Puram's hub-and-spoke, so Gate 3 has no hub to test. EMT-TFs monotone: ZEB1 (42% det) rises weakly to the terminus; canonical SNAI1/2, TWIST1, PRRX1, FOXC1 sit near the detection floor (no mesenchymal arm).

Parameter choices

Min cells/sample	≥269 (keeps 13/14; gap 561→129, r4.35)	**Integration**	Harmony(sample) θ=5; 50PC→20; UMAP spread=1.3
Malignant states	Noradren. progenitor 17,333 / Differentiating 10,654	**Trajectory**	scFates curve N=20 λ=0.3 μ=100 (fold-free, 2 tips)
Root node	tip 0 = Gast/pSom-high, Crest-low	**Manifold**	full malignant (no lobe split); 1 kNN component